Last time... Subspaces - 0 in subspace
closed under addition
closed under scalar mult.

Subspace = subset which is also a vector space.

Examples

$$\left.\begin{array}{l} \text{span}\{v_1, \dots, v_p\} \\ \text{Nul } A \\ \text{Col } A \end{array}\right\} \begin{array}{l} \\ A \in M_{m \times n} \\ \\ \end{array} \text{ subspaces}$$

Let $A = \begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix} \in M_{m \times n}$

$$\text{Col } A = \text{span}\{a_1, \dots, a_n\} \subseteq \mathbb{R}^m$$
$$= \{Av : v \in \mathbb{R}^n\}$$

Q. When is $\text{Col } A = \mathbb{R}^m$?

A. When every $b \in \mathbb{R}^m$ is equal to Av for some $v \in \mathbb{R}^n$.

i.e. $Av = b$

This is a system of linear equations and we can compute this.

Def. Suppose V and W are vector spaces.

A <u>linear transformation</u> is a function $T: V \to W$ such that

(i) $T(v+u) = T(v) + T(u) \quad \forall v, u \in V$

(ii) $T(cv) = cT(v) \quad \forall c \in \mathbb{R}$

Example 1 Fix $A \in M_{m \times n}$

Define $T(v) = Av$ $\quad v \in \mathbb{R}^n$

Check that $T: \mathbb{R}^n \to \mathbb{R}^m$ is a lin. trans.

Example

Let W = set of differentiable functions on $\mathbb{R}$

From calculus we know that this is a subspace of

V = set of all real-valued functions on $\mathbb{R}$

Define $D: W \to V$ by $D(f) = f'$

where f' is the derivative of $f \in W$.

From calculus we know

$$D(f+g) = (f+g)' = f' + g' = D(f) + D(g)$$

$$D(cf) = (cf)' = cf' \qquad \forall f, g \in W,\ c \in \mathbb{R}$$

$$= c\,D(f)$$

Def. The <u>null space</u> of a lin. trans $T: V \to W$

is $\text{Nul}\, T = \{ v \in V : T(v) = 0 \}$

The <u>range</u> of T is $\text{range}\, T = \{ T(v) : v \in V \}$

Exercise $\text{Nul}\, T$ and $\text{range}\, T$ are subspaces

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4.3 <u>Linearly independent sets and Bases</u>

In this section V is a (real) vector space and $H \subseteq V$ is a subspace.

Def. Suppose $v_1, \dots, v_p \in V$ ($\{v_1, \dots, v_p\} \subseteq V$).

We say these vectors a <u>linearly independent</u> if the only solution to

$$c_1 v_1 + \cdots + c_p v_p = 0 \qquad \text{with } c_1, \dots, c_p \in \mathbb{R}$$

is $c_1 = c_2 = \cdots = c_p = 0$.

If $\{v_1, \dots, v_p\}$ is not linearly independent we call it <u>linearly dependent</u>.

i.e. the exists $c_1, \dots, c_p \in \mathbb{R}$ such

$$\left.\begin{array}{l} c_1 v_1 + \cdots + c_p v_p = 0 \\ \text{and at least one } c_j \neq 0. \end{array}\right\} (*)$$

<u>linear dependence relation</u>

Example Suppose $v_1 = 0 \in V$.

Then $1 v_1 + 0 v_2 + \cdots + 0 v_p = 0 + 0 + \cdots + 0 = 0$

This shows that $\{0, v_2, \dots, v_p\}$ is lin. dep.

(i.e. $c_1 = 1$, $c_2 = \cdots = c_p = 0$ in $(*)$.

Theorem 4

Suppose $\{v_1, \dots, v_p\} \subseteq V$ with $p \geq 2$ and $v_1 \neq 0$.

Then $\{v_1, \dots, v_p\}$ is lin. dep. $\iff$ $v_j = d_1 v_1 + \cdots + d_{j-1} v_{j-1}$

for some $2 \leq j \leq p$

$d_1, \dots, d_{j-1} \in \mathbb{R}$

Example.

Suppose $\{v_1, v_2\} \subseteq V$ with $v_1 \neq 0$, is linearly dependent.

By Thm 4 $v_2 = d_1 v_1$ for some $d_1 \in \mathbb{R}$

This says that two vectors are linearly independent exactly when neither vector is a scalar multiple of the other.

Example

The functions $\{\sin, \cos\}$. We know that

$\cos \neq d \sin$ for any $d \in \mathbb{R}$

so by Thm 4 $\{\sin, \cos\}$ is lin. ind.

$\Big(1 = \cos(0) = d \sin(0) = d\,0 = 0 \Rightarrow d = 0.$

$\Rightarrow \cos = 0 \sin = 0$

contradiction$\Big)$

Example

In $\mathbb{P}$ define $p_1(x) = 1 + x + x^2$

$p_2(x) = 2 + x^2$

$p_3(x) = 2 + x + x^2 \quad \forall x \in \mathbb{R}.$

Is $\{p_1, p_2, p_3\}$ lin. ind.?

Suppose $c_1, c_2, c_3 \in \mathbb{R}$. Then

$$c_1 p_1(x) + c_2 p_2(x) + c_3 p_3(x)$$

$$= (c_1 + 2c_2 + 2c_3) + (c_1 + c_3)x + (c_1 + c_2 + c_3)x^2$$

$$= 0$$

exactly when $\begin{cases} c_1 + 2c_2 + 2c_3 = 0 \\ c_1 + c_3 = 0 \\ c_1 + c_2 + c_3 = 0 \end{cases}$ (**)

A system of linear equations. To solve this system row reduce $\begin{bmatrix} 1 & 2 & 2 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ to get $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

The solution to (**) is $c_1 = c_2 = c_2$.

Def. A <u>basis</u> for a subspace H is a set of vectors $\{v_1, \dots, v_p\} \subseteq H$ such that

(i) $\{v_1, \dots, v_p\}$ is lin. ind.

(ii) $\operatorname{span}\{v_1, \dots, v_p\} = H$.

Example $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a basis for $\mathbb{R}^3$.

In general $e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, e_n = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$

gives us the standard basis for $\mathbb{R}^n$.

Example $\mathbb{P}_1$ = polynomials of degree 1 or less.

Let $p_0(x) = 1$, $p_1(x) = x$ $\forall x \in \mathbb{R}$.

Then $\{p_0, p_1\}$ is a basis for $\mathbb{P}_1$.

Indeed $p_0 \neq 0$ and $p_1 \neq d p_0$ for any d.

($d p_0(x) = d(1) = d$, $p_1(x) = x$, $p_1(0) = 0 \neq d$ when $d \neq 0$)

So $\{p_0, p_1\}$ is lin. ind (see previous example.)

Furthermore if $p(x) = a_0 + a_1 x$ for $a_0, a_1 \in \mathbb{R}$

$$p = a_0 p_0 + a_1 p_1 \in \text{span}\{p_0, p_1\}.$$

This proves that $\mathbb{P}_1 = \text{span}\{p_0, p_1\}$.

Easier to just write $\{1, x\}$ is a basis for $\mathbb{P}_1$.

In general $\{1, x, x^2, \dots, x^n\}$ is a basis for $\mathbb{P}_n$.

(Exercise)

$\{1, x, x^2, \dots\}$ is a basis for $\mathbb{P}$.

Example Let $\{v_1, \dots, v_p\} \subseteq V$ $(v_1 \neq 0)$. Let $H = \text{span}\{v_1, \dots, v_p\}$.

H is a subspace. If $\{v_1, \dots, v_p\}$ is lin. ind. then it is a basis for H.

Otherwise $\{v_1, \dots, v_p\}$ is lin. dep and by Thm 4

$$v_j = d_1 v_1 + \dots + d_{j-1} v_{j-1} \quad \text{for some } 2 \leq j \leq p$$
$$d_1, \dots, d_{j-1} \in \mathbb{R}.$$

Thm 5

$$\text{span}\{v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_p\} = \text{span}\{v_1, \dots, v_p\}.$$

Idea. Discard v_j and check whether $\{v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_p\}$ is a basis for H.

Repeat this process until we find a basis.

Example

Let $H = \text{Span}\left\{ \underset{v_1}{\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}}, \underset{v_2}{\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}}, \underset{v_3}{\begin{bmatrix} 0 & 0 \\ -2 & -1 \end{bmatrix}} \right\} \subseteq M_{2\times 2}$.

To check whether $\{v_1, v_2, v_3\}$ is lin. ind./lin. dep

consider $c_1 v_1 + c_2 v_2 + c_3 v_3 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

This reduces to $c_1 \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 0 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

or $\begin{bmatrix} 2c_1 + c_2 & 0 \\ c_2 - 2c_3 & c_1 + c_2 - c_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

The solutions here are solutions to the system

$$\begin{cases} 2c_1 + c_2 = 0 \\ c_2 - 2c_3 = 0 \\ c_1 + c_2 - c_3 = 0 \end{cases}$$

Row reduce $\begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$